

System Model

Design Requirements

Voltage

Current

Design Variables

Switching Frequency

Common-mode Inductance

Filter Inductance

Analytical Model

Efficiency Model

- * Power Component Loss
- * Magnetic Component Loss

Volume Model

- * Filter Inductor Volume
- * Common-mode Inductor Volume

Cost Model

- * Core Cost
- * Winding Cost

Ripple Model

ANN Model

CM Noise Model



ANN_C Magnitude



ANN_S Frequency

Design Objectives

Efficiency

Volume

Cost

Ripple

Common-mode Noise